

Ayush Mani Nepal



PROFILE

Software and Tech enthusiast with a passion for problem-solving is seeking a new opportunity

CONTACT

✉ aaus_nep@hotmail.com
🌐 ayushnepal.github.io
📄 ayushnepal
🏠 38442 Wolfsburg, Germany

PERSONAL INFO

BIRTH YEAR: 1990

LANGUAGES:

- ◇ German (Business fluent)
- ◇ English (Business fluent)
- ◇ Nepali (Mother-tongue)

MAIN COMPETENCIES

- ◇ Team-spirit, Communication
- ◇ Software (Design, Architecture, Development, Operation)
- ◇ SW-Documentation (Concept, Architecture, Detailed Design)
- ◇ Agile, SOLID, Clean-Code
- ◇ C++, Java, C#, Python
- ◇ SW-Model, MBSE, MDD
- ◇ UML, XText, Xtend, Code-gen

TECHNICAL KNOW-HOW

- ◇ Linux, Windows, Android
- ◇ Git, Git-lab, SVN, CI/CD
- ◇ MVC, MVVC, Observer, MPI, etc.
- ◇ Testing, Test Automation, TDD
- ◇ Deep Learning, PyTorch, CUDA
- ◇ Java-script, HTML, CSS
- ◇ Database, SQL, Container
- ◇ Azure, JIRA, Confluence

HOBBIES

Music, Traveling, Hiking

PROFESSIONAL EXPERIENCES

SENIOR SOFTWARE ENGINEER FULL-STACK

IAV GmbH

Nov 2024 — Now
Gifhorn, Germany

- Designed and developed data-structures and algorithm to enable seamless linking between DAB and FM alternatives dynamically
- Designed and developed data-structures and algorithm to efficiently maintain list of receivable stations

MIDDLEWARE SOFTWARE ENGINEER

IAV GmbH

Jun 2023 — Nov 2024
Gifhorn, Germany

- Contributed significantly in the development of middleware software for a radio-tuner in C++

R&D SCIENTIST

German Aerospace Center (DLR)

Jun 2019 — May 2023
Braunschweig, Germany

- Designed and developed MBSE toolsets for various space missions to facilitate model-driven development of the flight software ↗
- Researched and developed machine learning models for their applications in space systems ↗
- Contributed to an open-source MBSE tool for model-based software development ↗
- Supervised students in their thesis projects and internships

MASTER THESIS

German Aerospace Center (DLR)

Nov 2018 — Apr 2019
Braunschweig, Germany

- Modeling and Simulation of Spacecraft Payload Hardware using Machine Learning Techniques ↗

STUDENT JOB

German Aerospace Center (DLR)

May 2017 — Apr 2019
Braunschweig, Germany

- Model-based system engineering for space systems

ELECTRICAL AND IT ENGINEER

Airbus Defence and Space (extern)

Apr 2015 — Sep 2016
Immenstaad, Germany

- Designed and developed test-automation framework to test the Off-Board Map Preparation System for fighter planes
- Automated thousands of software and system test-cases using the self-developed framework

ENGINEERING CONSULTANT

ALTEN GmbH

Nov 2014 — Mar 2015
Munich, Germany

- Front-end development in WPF/C# for the Building-automation

STUDENT JOB

Airbus Defence and Space

Oct 2012 — Dec 2013
Ulm, Germany

- Application development in C#/.Net and C++

COURSES AND CERTIFICATIONS

- ◇ ISAQB CERTIFIED PROFESSIONAL FOR SOFTWARE ARCHITECTURE ↗
- ◇ CLEAN-CODE DEVELOPER ASSOCIATE FROM CODELUTION ↗
- ◇ SYSTEMS ENGINEERING FROM UNITY CONSULTING & INNOVATION
- ◇ PROJECT MANAGEMENT COMPACT

STUDIES

M.SC. COMPUTATIONAL SCIENCES IN ENGINEERING (CSE)

TU Braunschweig

2017 — 2019
Braunschweig, Germany

B.E. ELECTRICAL ENGINEERING AND INFO TECH

FH Ravensburg-Weingarten

2010 — 2014
Weingarten, Germany